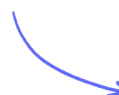


Bearings

Bearings

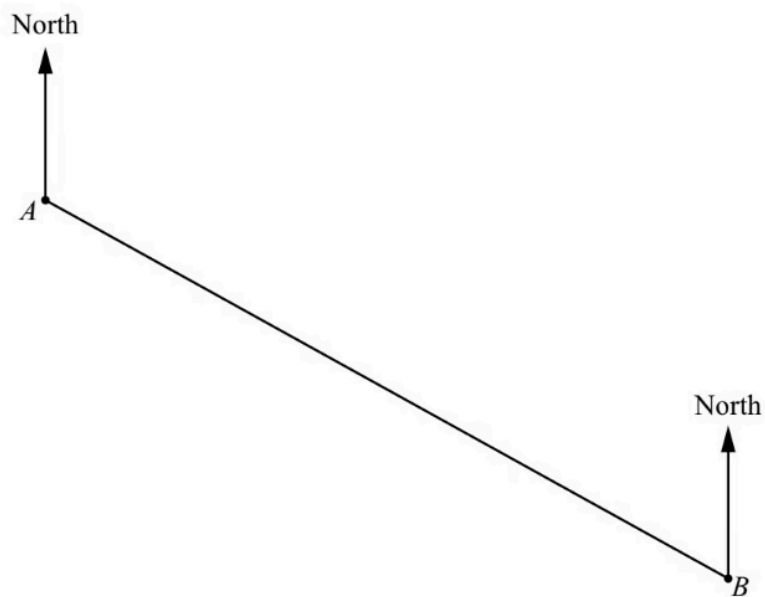
Easy (5 questions)	/9
Medium (8 questions)	/19
Hard (10 questions)	/51
Total Marks	/79

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Easy Questions

1



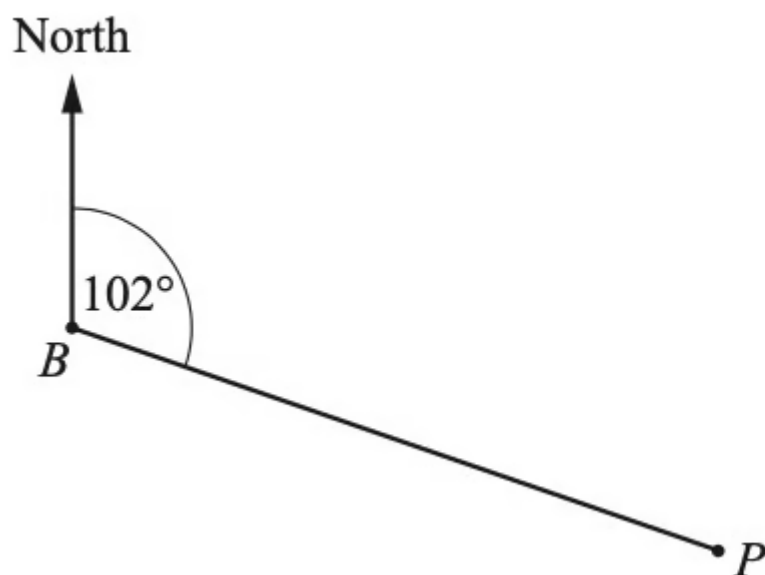
Two towns, *A* and *B*, are shown on a map.
The scale of the map is 1 cm to 3 km.

Find the actual distance between *A* and *B*.

..... km

(1 mark)

2



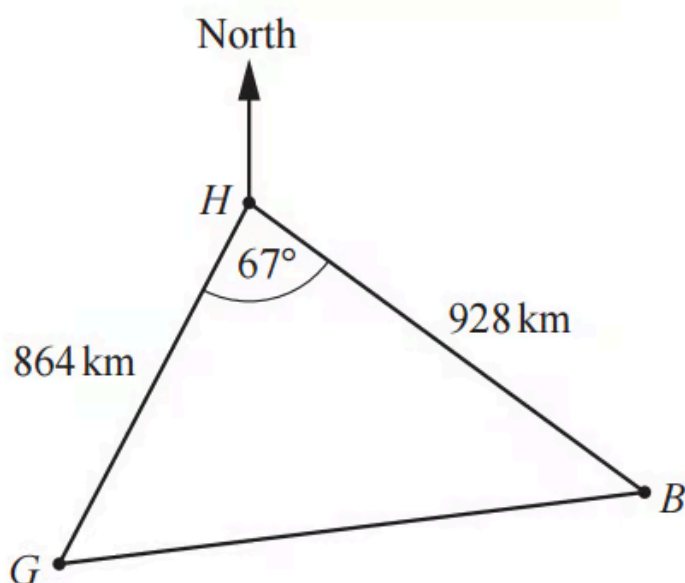
NOT TO
SCALE

The bearing of P from B is 102° .

Find the bearing of B from P .

(2 marks)

- 3 The diagram shows the positions of three cities, Geneva (G), Budapest (B) and Hamburg (H).



NOT TO
SCALE

The bearing of Budapest from Hamburg is 133° .

Find the bearing of Hamburg from Budapest.

(2 marks)

- 4 The bearing of a ship from a lighthouse is 050°

Work out the bearing of the lighthouse from the ship.

(2 marks)

- 5 Martin and Janet are in an orienteering race.

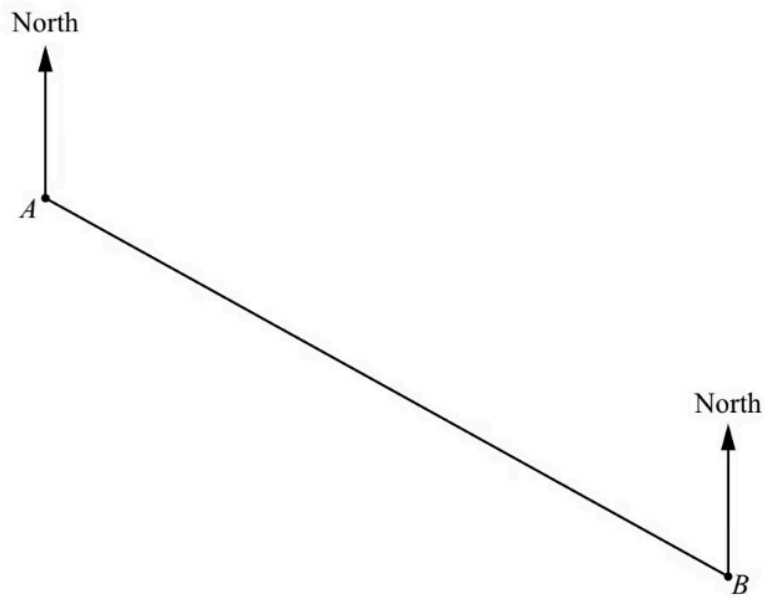
Martin runs from checkpoint A to checkpoint B , on a bearing of 065° . Janet is going to run from checkpoint B to checkpoint A .

Work out the bearing of A from B .

(2 marks)

Medium Questions

1 (a)



Two towns, A and B , are shown on a map.

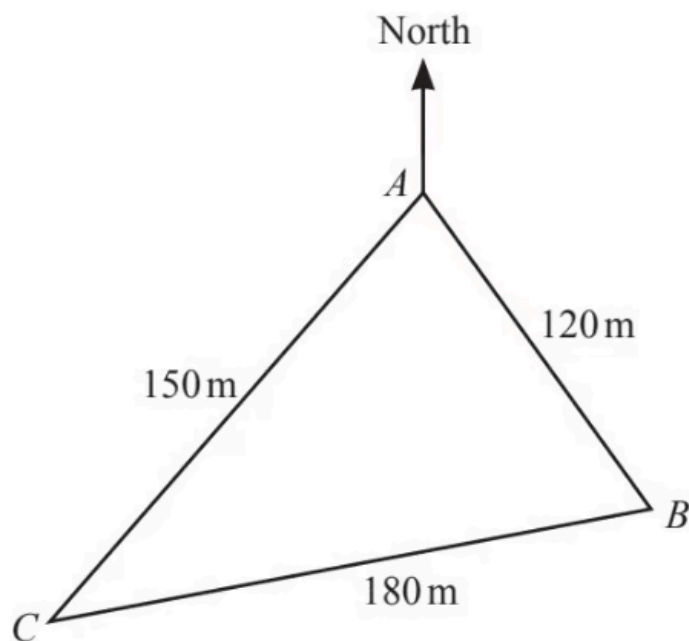
Measure the bearing of B from A .

(1 mark)

- (b) Calculate the bearing of A from B .
You must show all your working.

(2 marks)

2



NOT TO
SCALE

The diagram shows a triangular field, ABC , on horizontal ground.

The angle BAC is equal to 82.8° .

The bearing of C from A is 210° .

i) Find the bearing of B from A .

[1]

ii) Find the bearing of A from B .

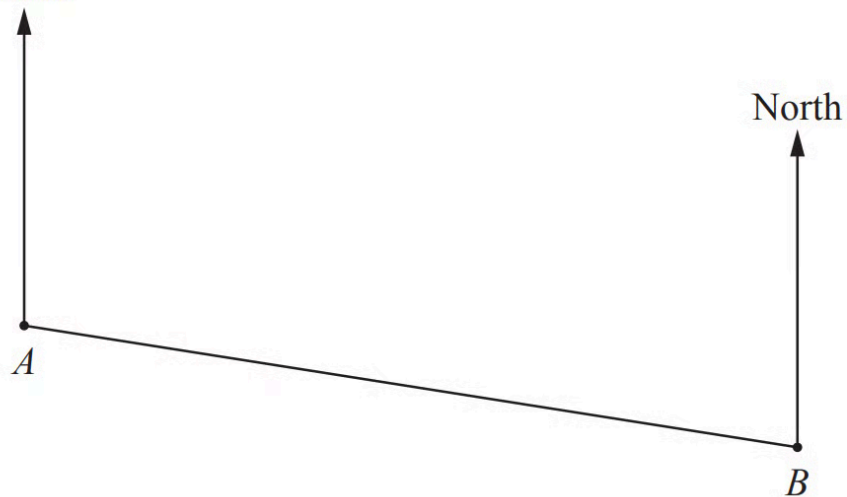
[2]

(3 marks)

3

North

NOT TO
SCALE

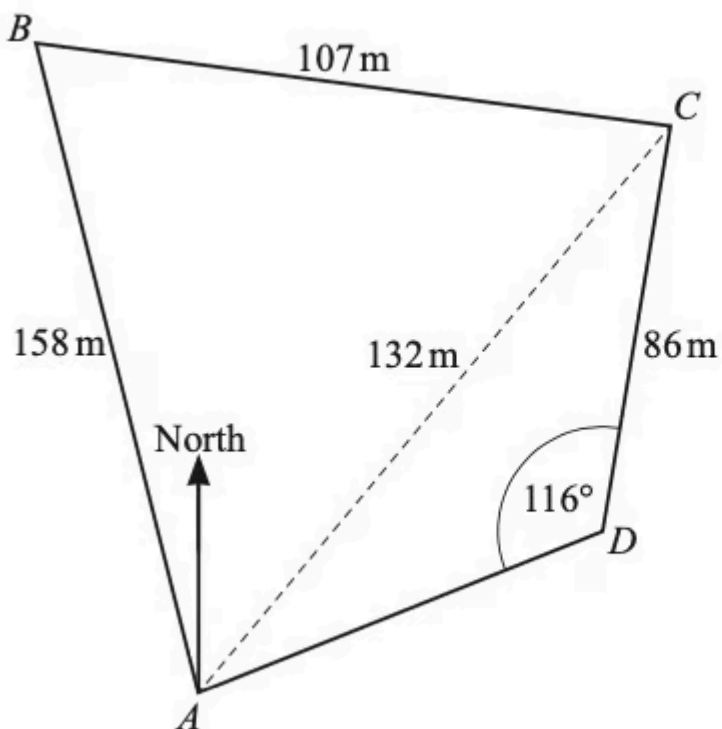


The bearing of B from A is 105° .

Find the bearing of A from B .

(2 marks)

4



NOT TO
SCALE

The diagram shows a field, $ABCD$, on horizontal ground.

The angle CAD is 35.8° and the bearing of D from A is 070° .

Find the bearing of A from C .

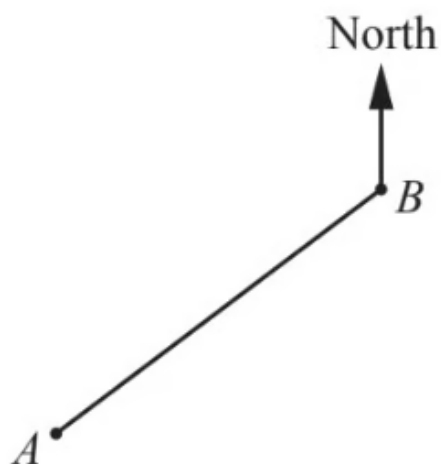
(2 marks)

5 The bearing of Alexandria from Paris is 128° .

Calculate the bearing of Paris from Alexandria.

(2 marks)

6



NOT TO
SCALE

The bearing of A from B is 227° .

Find the bearing of B from A .

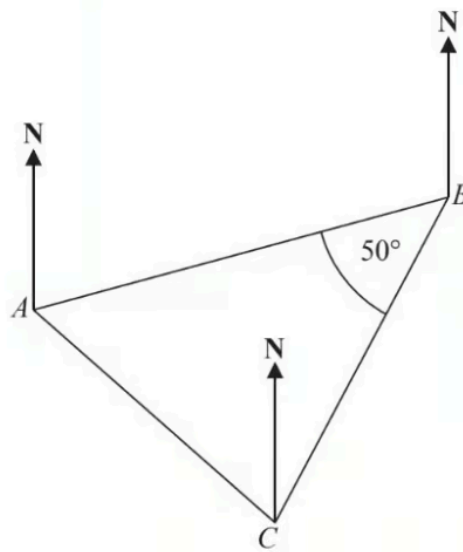
(2 marks)

- 7 A and B are two towns on a map.
The bearing of A from B is 140° .

Work out the bearing of B from A .

(2 marks)

- 8 The diagram shows the positions of three points, A , B and C , on a map.



The bearing of B from A is 070° .

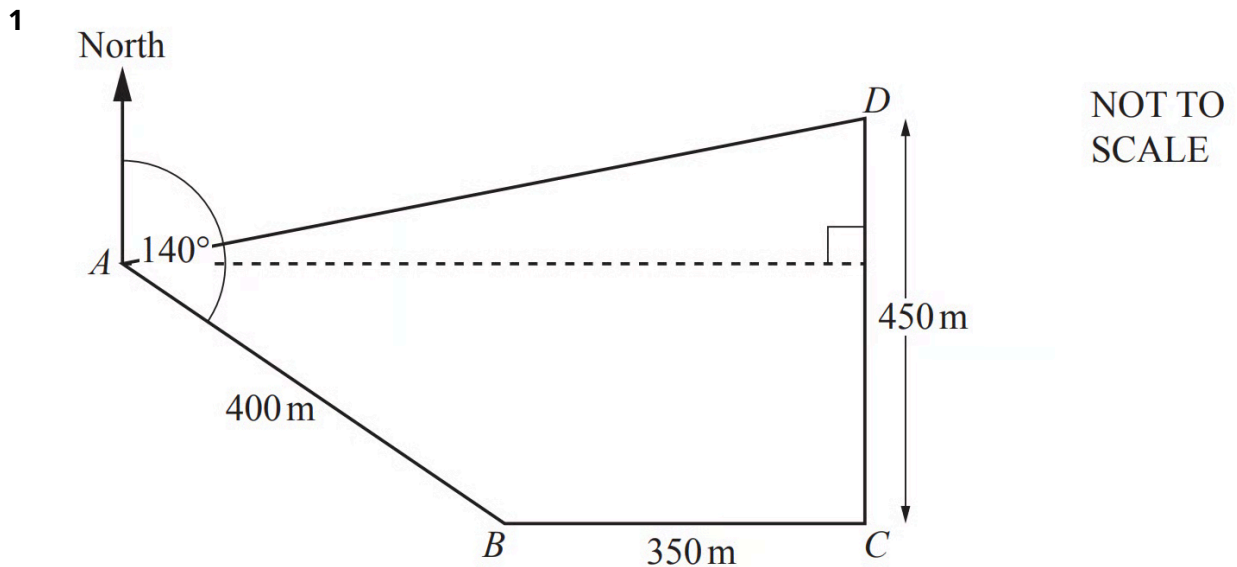
Angle ABC is 50° .

$AB = CB$

Work out the bearing of C from A .

(3 marks)

Hard Questions



The diagram shows a field $ABCD$.

The bearing of B from A is 140° .

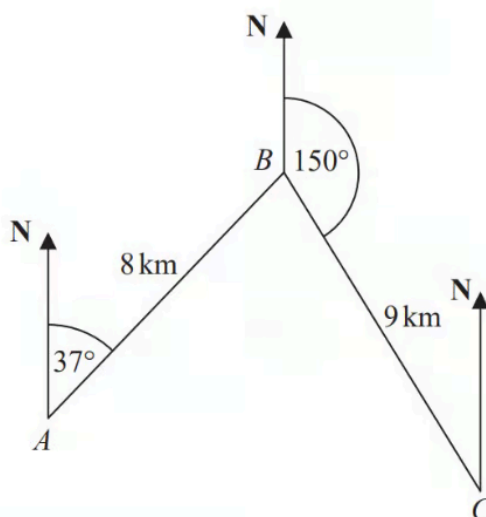
C is due east of B and D is due north of C .

$AB = 400\text{m}$, $BC = 350\text{m}$ and $CD = 450\text{m}$.

Find the bearing of D from B .

(2 marks)

- 2 The diagram shows the positions of three towns, Acton (A), Barston (B) and Chorlton (C).



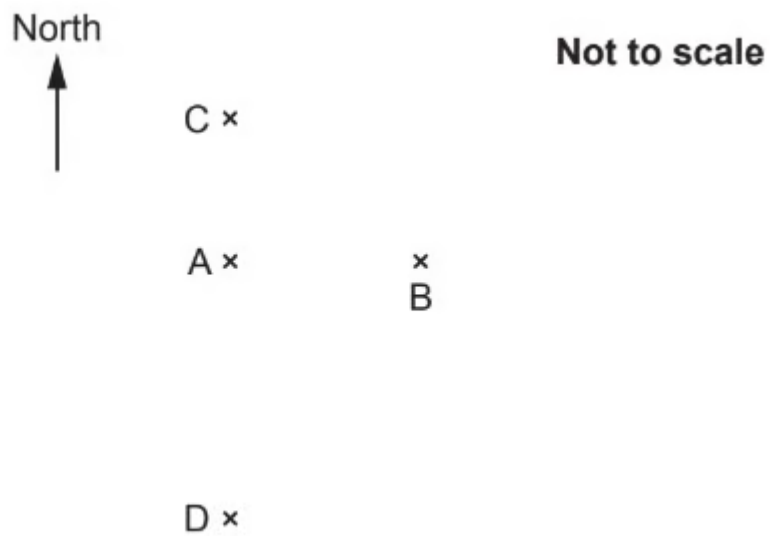
Barston is 8 km from Acton on a bearing of 037° . Chorlton is 9 km from Barston on a bearing of 150° .

Find the bearing of Chorlton from Acton.
Give your answer correct to 1 decimal place.
You must show all your working.

(5 marks)

- 3 (a) A, B, C and D are four towns.

B is 25 kilometres due East of A.
C is 25 kilometres due North of A.
D is 45 kilometres due South of A.



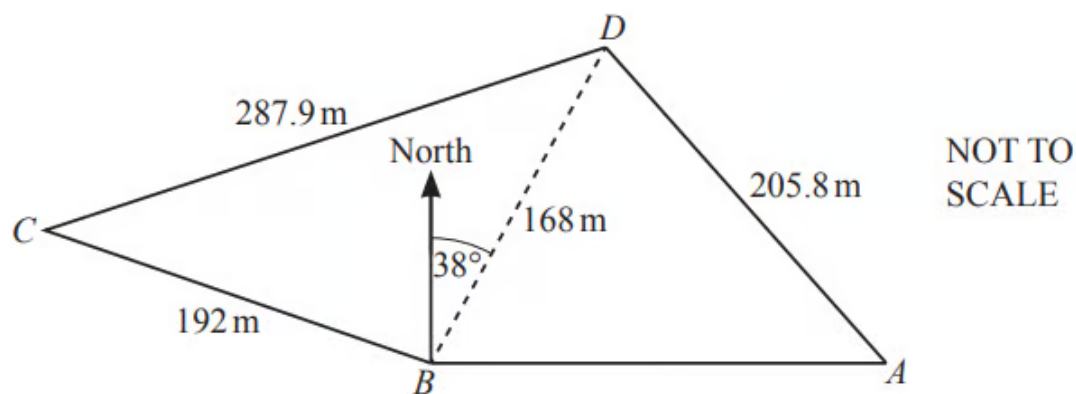
Work out the bearing of B from C.

(2 marks)

(b) Calculate the bearing of D from B.

(4 marks)

4



The diagram shows a field, $ABCD$, on horizontal ground. $BC = 192$ m, $CD = 287.9$ m, $BD = 168$ m and $AD = 205.8$ m.

Angle $CBD = 106^\circ$.

i) The bearing of D from B is 038° .

Find the bearing of C from B .

[1]

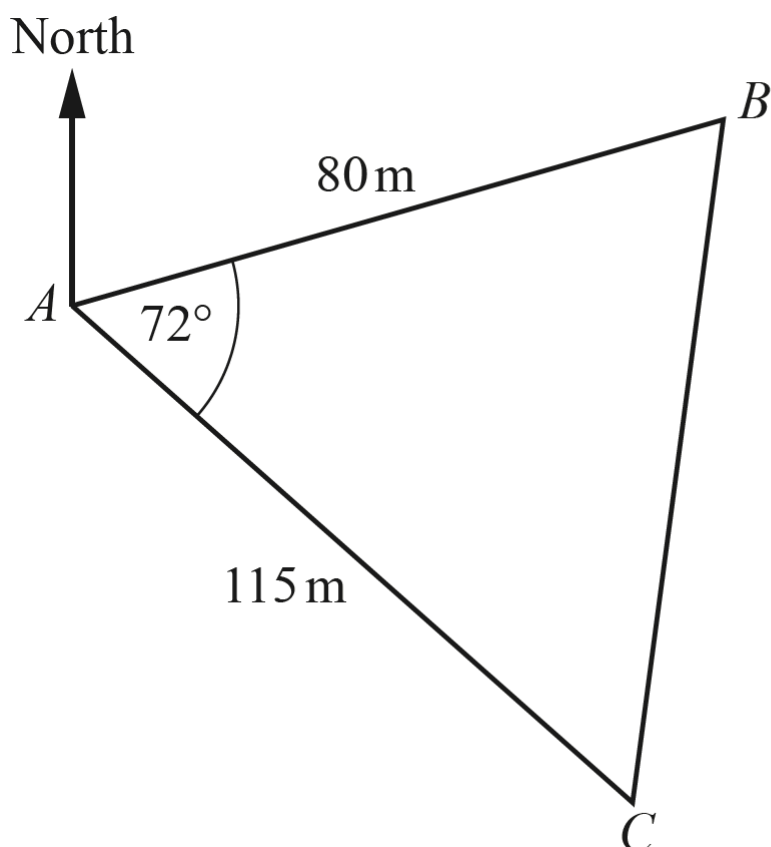
ii) A is **due east** of B .

Calculate the bearing of D from A .

[5]

(6 marks)

5 (a)



NOT TO
SCALE

The diagram shows the positions of three points A , B and C in a field.

Show that BC is 118.1 m, correct to 1 decimal place.

(3 marks)

(b) Calculate angle ABC .

Angle ABC =

(3 marks)

(c) The bearing of C from A is 147° . Find the bearing of

i) A from B ,

[3]

ii) B from C .

[2]

(5 marks)

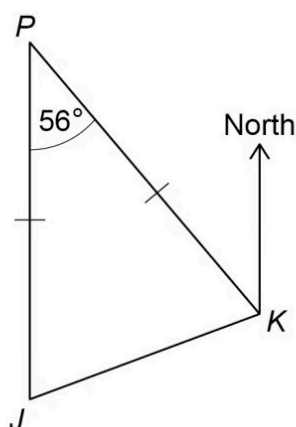
6 J and K are ships.

P is a port.

J is due South of P .

Angle $JPK = 56^\circ$

$JP = KP$

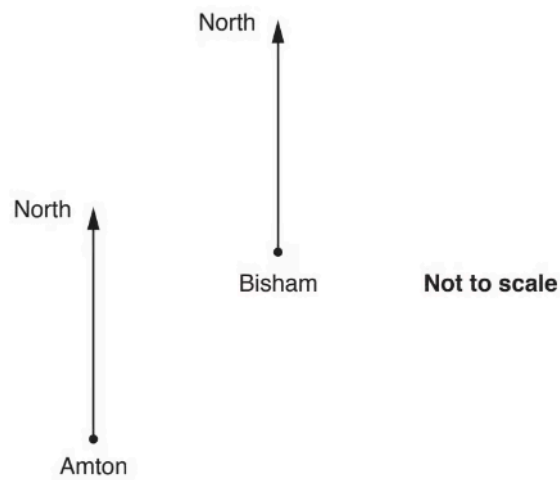


Not drawn
accurately

Work out the bearing of J from K .

(3 marks)

- 7 The diagram shows the positions of two towns, Amton and Bisham.



The bearing of Bisham from Amton is b° .

The bearing of Amton from Bisham is $6b^\circ$.

Calculate the 3-figure bearing of Amton from Bisham.

(4 marks)

- 8 The diagram shows the positions of three ships, A , B and C .

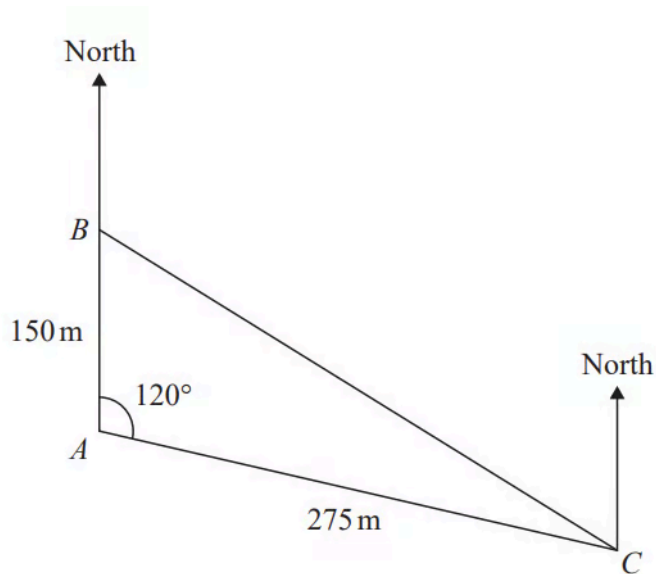


Diagram **NOT**
accurately drawn

Ship B is due north of ship A .

The bearing of ship C from ship A is 120°

Calculate the bearing of ship C from ship B .
Give your answer correct to the nearest degree.

(5 marks)

- 9 A , B and C are three towns.

The bearing of B from A is 105°

The bearing of C from B is 230°

The distance of C from A is 180 km.

The distance of C from B is 95 km.

Calculate the distance of B from A .
 Give your answer correct to 3 significant figures.

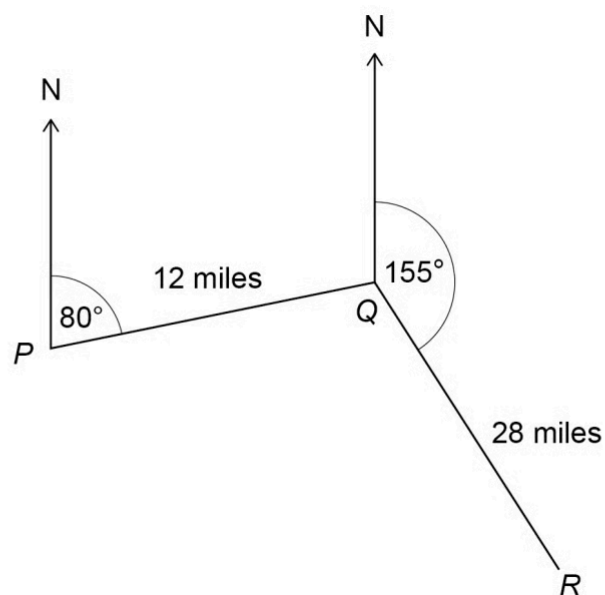
.....km

(5 marks)

10 A ship sails from P to Q and then from Q to R .

Q is 12 miles from P , on a bearing of 080°

R is 28 miles from Q , on a bearing of 155°



Work out the direct distance from P to R .

(4 marks)